# Title : Data Wrangling on Real Estate Market

In [2]:
```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import warnings
```

In [3]:
```
# Supressing update warnings
warnings.filterwarnings('ignore')
```

In [4]:
```
df1 = pd.read_csv("./datasets/Bengaluru_House_Data.csv")
```

In [5]:
```
df1.head()
```

Out[5]:

| | area_type | availability | location | size | society | total_sqft | bath | balco |
|---|---|---|---|---|---|---|---|---|
| **0** | Super built-up Area | 19-Dec | Electronic City Phase II | 2 BHK | Coomee | 1056 | 2.0 | |
| **1** | Plot Area | Ready To Move | Chikka Tirupathi | 4 Bedroom | Theanmp | 2600 | 5.0 | |
| **2** | Built-up Area | Ready To Move | Uttarahalli | 3 BHK | NaN | 1440 | 2.0 | |
| **3** | Super built-up Area | Ready To Move | Lingadheeranahalli | 3 BHK | Soiewre | 1521 | 3.0 | |
| **4** | Super built-up Area | Ready To Move | Kothanur | 2 BHK | NaN | 1200 | 2.0 | |

In [6]:
```
df1.shape
```

Out[6]:
```
(13320, 9)
```

In [7]:
```
df1.columns
```

Out[7]:
```
Index(['area_type', 'availability', 'location', 'size', 'society',
       'total_sqft', 'bath', 'balcony', 'price'],
      dtype='object')
```

In [8]:
```
df1['area_type']
```

Out[8]:
```
0        Super built-up  Area
1                  Plot  Area
2              Built-up  Area
3        Super built-up  Area
4        Super built-up  Area
                 ...
13315          Built-up  Area
13316    Super built-up  Area
13317          Built-up  Area
13318    Super built-up  Area
13319    Super built-up  Area
Name: area_type, Length: 13320, dtype: object
```

In [9]:
```
df1['area_type'].unique()
```

Out[9]:
```
array(['Super built-up  Area', 'Plot  Area', 'Built-up  Area',
       'Carpet  Area'], dtype=object)
```

In [10]:
```
df1['area_type'].value_counts()
```

Out[10]:
```
Super built-up  Area    8790
Built-up  Area          2418
Plot  Area              2025
Carpet  Area              87
Name: area_type, dtype: int64
```

In [11]:
```
df2 = df1.drop(['area_type','society','balcony','availability'],axis='columns')
```

In [12]:
```
df2.shape
```

Out[12]:
```
(13320, 5)
```

In [13]:
```
df2.isnull().sum()
```

Out[13]:
```
location       1
size          16
total_sqft     0
bath          73
price          0
dtype: int64
```

In [14]:
```
df2.shape
```

Out[14]:
```
(13320, 5)
```

In [15]:
```
df3 = df2.dropna()
df3.isnull().sum()
```

Out[15]:
```
location      0
size          0
total_sqft    0
bath          0
price         0
dtype: int64
```

In [16]:
```
df3.shape
```

Out[16]:
```
(13246, 5)
```

In [17]:
```
df3['size'].unique()
```

Out[17]:
```
array(['2 BHK', '4 Bedroom', '3 BHK', '4 BHK', '6 Bedroom', '3 Bedroom',
       '1 BHK', '1 RK', '1 Bedroom', '8 Bedroom', '2 Bedroom',
       '7 Bedroom', '5 BHK', '7 BHK', '6 BHK', '5 Bedroom', '11 BHK',
       '9 BHK', '9 Bedroom', '27 BHK', '10 Bedroom', '11 Bedroom',
       '10 BHK', '19 BHK', '16 BHK', '43 Bedroom', '14 BHK', '8 BHK',
       '12 Bedroom', '13 BHK', '18 Bedroom'], dtype=object)
```

In [18]:
```
df3['bhk'] = df3['size'].apply(lambda x: int(x.split(' ')[0]))
```

In [19]:
```
df3.head()
```

Out[19]:

| | location | size | total_sqft | bath | price | bhk |
|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056 | 2.0 | 39.07 | 2 |
| **1** | Chikka Tirupathi | 4 Bedroom | 2600 | 5.0 | 120.00 | 4 |
| **2** | Uttarahalli | 3 BHK | 1440 | 2.0 | 62.00 | 3 |
| **3** | Lingadheeranahalli | 3 BHK | 1521 | 3.0 | 95.00 | 3 |
| **4** | Kothanur | 2 BHK | 1200 | 2.0 | 51.00 | 2 |

In [20]:
```
df3.bhk.unique()
```

Out[20]:
```
array([ 2,  4,  3,  6,  1,  8,  7,  5, 11,  9, 27, 10, 19, 16, 43, 14, 12,
       13, 18], dtype=int64)
```

In [21]:
```
df3[df3.bhk>20]
```

Out[21]:

| | location | size | total_sqft | bath | price | bhk |
|---|---|---|---|---|---|---|
| **1718** | 2Electronic City Phase II | 27 BHK | 8000 | 27.0 | 230.0 | 27 |
| **4684** | Munnekollal | 43 Bedroom | 2400 | 40.0 | 660.0 | 43 |

In [22]:
```
df3.total_sqft.unique()
```

Out[22]:
```
array(['1056', '2600', '1440', ..., '1133 - 1384', '774', '4689'],
      dtype=object)
```

In [23]:
```
def is_float(x):
    try:
        float(x)
        return True
    except(ValueError, TypeError):
        return False
```

In [24]:
```
df3[~df3['total_sqft'].apply(is_float)].head(10)
```

Out[24]:

| | location | size | total_sqft | bath | price | bhk |
|---|---|---|---|---|---|---|
| **30** | Yelahanka | 4 BHK | 2100 - 2850 | 4.0 | 186.000 | 4 |
| **122** | Hebbal | 4 BHK | 3067 - 8156 | 4.0 | 477.000 | 4 |
| **137** | 8th Phase JP Nagar | 2 BHK | 1042 - 1105 | 2.0 | 54.005 | 2 |
| **165** | Sarjapur | 2 BHK | 1145 - 1340 | 2.0 | 43.490 | 2 |
| **188** | KR Puram | 2 BHK | 1015 - 1540 | 2.0 | 56.800 | 2 |
| **410** | Kengeri | 1 BHK | 34.46Sq. Meter | 1.0 | 18.500 | 1 |
| **549** | Hennur Road | 2 BHK | 1195 - 1440 | 2.0 | 63.770 | 2 |
| **648** | Arekere | 9 Bedroom | 4125Perch | 9.0 | 265.000 | 9 |
| **661** | Yelahanka | 2 BHK | 1120 - 1145 | 2.0 | 48.130 | 2 |
| **672** | Bettahalsoor | 4 Bedroom | 3090 - 5002 | 4.0 | 445.000 | 4 |

In [25]:
```
def convert_sqft_to_num(x):
    tokens = x.split('-')
    if len(tokens) == 2:
        try:
            return (float(tokens[0])+float(tokens[1]))/2
        except ValueError:
            return None
    try:
        return float(x)
    except ValueError:
        return None

result = convert_sqft_to_num('2100 - 2850')
print(result)
```

```
2475.0
```

In [26]:
```
convert_sqft_to_num('34.46Sq. Meter')
df4 = df3.copy()
df4.total_sqft = df4.total_sqft.apply(convert_sqft_to_num)
df4
```

Out[26]:

| | location | size | total_sqft | bath | price | bhk |
|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056.0 | 2.0 | 39.07 | 2 |
| **1** | Chikka Tirupathi | 4 Bedroom | 2600.0 | 5.0 | 120.00 | 4 |
| **2** | Uttarahalli | 3 BHK | 1440.0 | 2.0 | 62.00 | 3 |
| **3** | Lingadheeranahalli | 3 BHK | 1521.0 | 3.0 | 95.00 | 3 |
| **4** | Kothanur | 2 BHK | 1200.0 | 2.0 | 51.00 | 2 |
| **...** | ... | ... | ... | ... | ... | ... |
| **13315** | Whitefield | 5 Bedroom | 3453.0 | 4.0 | 231.00 | 5 |
| **13316** | Richards Town | 4 BHK | 3600.0 | 5.0 | 400.00 | 4 |
| **13317** | Raja Rajeshwari Nagar | 2 BHK | 1141.0 | 2.0 | 60.00 | 2 |
| **13318** | Padmanabhanagar | 4 BHK | 4689.0 | 4.0 | 488.00 | 4 |
| **13319** | Doddathoguru | 1 BHK | 550.0 | 1.0 | 17.00 | 1 |

13246 rows × 6 columns

In [27]:
```
df4 = df4[df4.total_sqft.notnull()]
df4
```

Out[27]:

| | location | size | total_sqft | bath | price | bhk |
|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056.0 | 2.0 | 39.07 | 2 |
| **1** | Chikka Tirupathi | 4 Bedroom | 2600.0 | 5.0 | 120.00 | 4 |
| **2** | Uttarahalli | 3 BHK | 1440.0 | 2.0 | 62.00 | 3 |
| **3** | Lingadheeranahalli | 3 BHK | 1521.0 | 3.0 | 95.00 | 3 |
| **4** | Kothanur | 2 BHK | 1200.0 | 2.0 | 51.00 | 2 |
| **...** | ... | ... | ... | ... | ... | ... |
| **13315** | Whitefield | 5 Bedroom | 3453.0 | 4.0 | 231.00 | 5 |
| **13316** | Richards Town | 4 BHK | 3600.0 | 5.0 | 400.00 | 4 |
| **13317** | Raja Rajeshwari Nagar | 2 BHK | 1141.0 | 2.0 | 60.00 | 2 |
| **13318** | Padmanabhanagar | 4 BHK | 4689.0 | 4.0 | 488.00 | 4 |
| **13319** | Doddathoguru | 1 BHK | 550.0 | 1.0 | 17.00 | 1 |

13200 rows × 6 columns

In [28]:
```
df4.loc[30]
```

Out[28]:
```
location      Yelahanka
size              4 BHK
total_sqft       2475.0
bath                4.0
price             186.0
bhk                   4
Name: 30, dtype: object
```

In [29]:
```
df5 = df4.copy()
df5['price_per_sqft'] = df5['price']*100000/df5['total_sqft']
df5.head()
```

Out[29]:

| | location | size | total_sqft | bath | price | bhk | price_per_sqft |
|---|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056.0 | 2.0 | 39.07 | 2 | 3699.810606 |
| **1** | Chikka Tirupathi | 4 Bedroom | 2600.0 | 5.0 | 120.00 | 4 | 4615.384615 |
| **2** | Uttarahalli | 3 BHK | 1440.0 | 2.0 | 62.00 | 3 | 4305.555556 |
| **3** | Lingadheeranahalli | 3 BHK | 1521.0 | 3.0 | 95.00 | 3 | 6245.890861 |
| **4** | Kothanur | 2 BHK | 1200.0 | 2.0 | 51.00 | 2 | 4250.000000 |

In [30]:
```
df5_stats = df5['price_per_sqft'].describe()
df5_stats
```

Out[30]:
```
count    1.320000e+04
mean     7.920759e+03
std      1.067272e+05
min      2.678298e+02
25%      4.267701e+03
50%      5.438331e+03
75%      7.317073e+03
max      1.200000e+07
Name: price_per_sqft, dtype: float64
```

In [31]:
```
df5.to_csv("./datasets/bhp.csv",index=False)
```

In [32]:
```
df5.location = df5.location.apply(lambda x: x.strip())
location_stats = df5['location'].value_counts(ascending=False)
location_stats
```

Out[32]:
```
Whitefield                    533
Sarjapur  Road                392
Electronic City               304
Kanakpura Road                264
Thanisandra                   235
                             ...
Rajanna Layout                  1
Subramanyanagar                 1
Lakshmipura Vidyaanyapura       1
Malur Hosur Road                1
Abshot Layout                   1
Name: location, Length: 1287, dtype: int64
```

In [33]:
```
len(location_stats[location_stats>10])
```

Out[33]:
```
240
```

In [34]:
```
len(location_stats)
```

Out[34]:
```
1287
```

In [35]:
```
len(location_stats[location_stats<=10])
```

Out[35]:
```
1047
```

In [36]:
```
location_stats_less_than_10 = location_stats[location_stats<=10]
location_stats_less_than_10
```

Out[36]:
```
BTM 1st Stage               10
Gunjur Palya                10
Nagappa Reddy Layout        10
Sector 1 HSR Layout         10
Thyagaraja Nagar            10
                            ..
Rajanna Layout               1
Subramanyanagar              1
Lakshmipura Vidyaanyapura    1
Malur Hosur Road             1
Abshot Layout                1
Name: location, Length: 1047, dtype: int64
```

In [37]:
```
len(df5.location.unique())
```

Out[37]:
```
1287
```

In [38]:
```
df5.location = df5.location.apply(lambda x: 'other' if x in location_stats_less_
len(df5.location.unique())
```

Out[38]:
```
241
```

In [39]:
```
df5.head(10)
```

Out[39]:

| | location | size | total_sqft | bath | price | bhk | price_per_sqft |
|---|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056.0 | 2.0 | 39.07 | 2 | 3699.810606 |
| **1** | Chikka Tirupathi | 4 Bedroom | 2600.0 | 5.0 | 120.00 | 4 | 4615.384615 |
| **2** | Uttarahalli | 3 BHK | 1440.0 | 2.0 | 62.00 | 3 | 4305.555556 |
| **3** | Lingadheeranahalli | 3 BHK | 1521.0 | 3.0 | 95.00 | 3 | 6245.890861 |
| **4** | Kothanur | 2 BHK | 1200.0 | 2.0 | 51.00 | 2 | 4250.000000 |
| **5** | Whitefield | 2 BHK | 1170.0 | 2.0 | 38.00 | 2 | 3247.863248 |
| **6** | Old Airport Road | 4 BHK | 2732.0 | 4.0 | 204.00 | 4 | 7467.057101 |
| **7** | Rajaji Nagar | 4 BHK | 3300.0 | 4.0 | 600.00 | 4 | 18181.818182 |
| **8** | Marathahalli | 3 BHK | 1310.0 | 3.0 | 63.25 | 3 | 4828.244275 |
| **9** | other | 6 Bedroom | 1020.0 | 6.0 | 370.00 | 6 | 36274.509804 |

In [40]:
```
df5[df5.total_sqft/df5.bhk<300].head()
```

Out[40]:

| | location | size | total_sqft | bath | price | bhk | price_per_sqft |
|---|---|---|---|---|---|---|---|
| **9** | other | 6 Bedroom | 1020.0 | 6.0 | 370.0 | 6 | 36274.509804 |
| **45** | HSR Layout | 8 Bedroom | 600.0 | 9.0 | 200.0 | 8 | 33333.333333 |
| **58** | Murugeshpalya | 6 Bedroom | 1407.0 | 4.0 | 150.0 | 6 | 10660.980810 |
| **68** | Devarachikkanahalli | 8 Bedroom | 1350.0 | 7.0 | 85.0 | 8 | 6296.296296 |
| **70** | other | 3 Bedroom | 500.0 | 3.0 | 100.0 | 3 | 20000.000000 |

In [41]:
```
df5.shape
```

Out[41]:
```
(13200, 7)
```

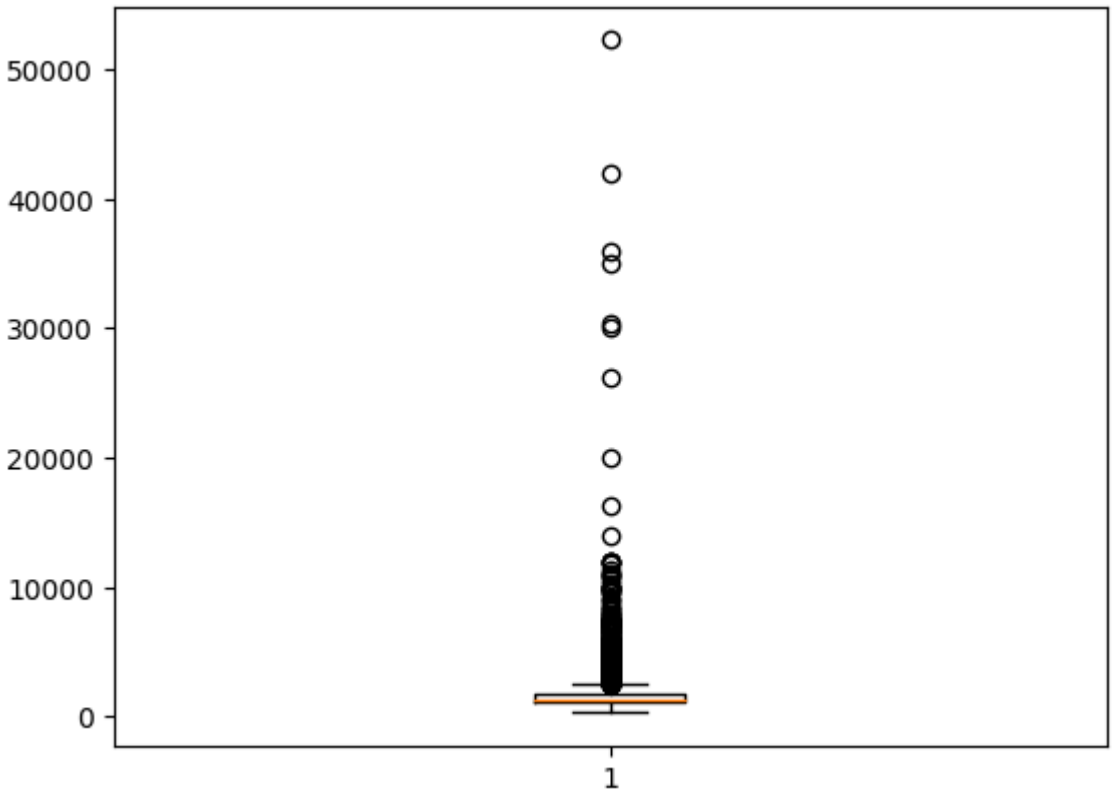
In [42]:
```
df6 = df5[~(df5.total_sqft/df5.bhk<300)]
df6.shape
```

Out[42]:
```
(12456, 7)
```

In [43]:
```
df6.columns
```

Out[43]:
```
Index(['location', 'size', 'total_sqft', 'bath', 'price', 'bhk',
       'price_per_sqft'],
      dtype='object')
```

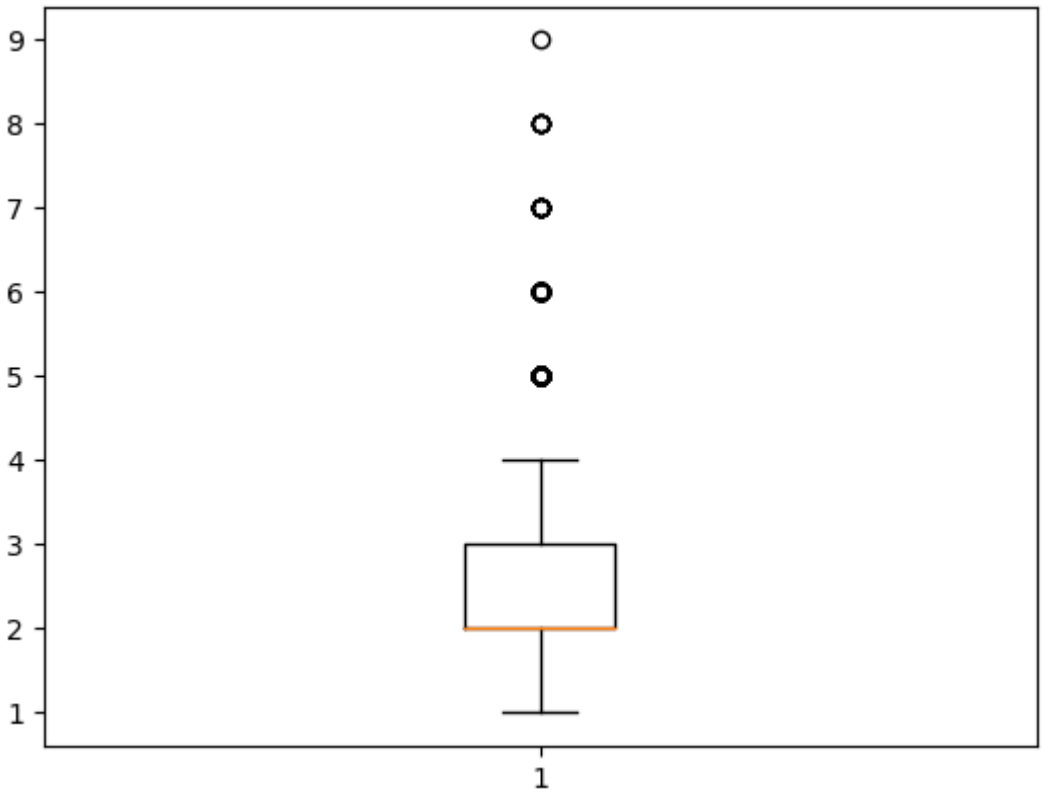
In [44]:
```
plt.boxplot(df6['total_sqft'])
plt.show()
```



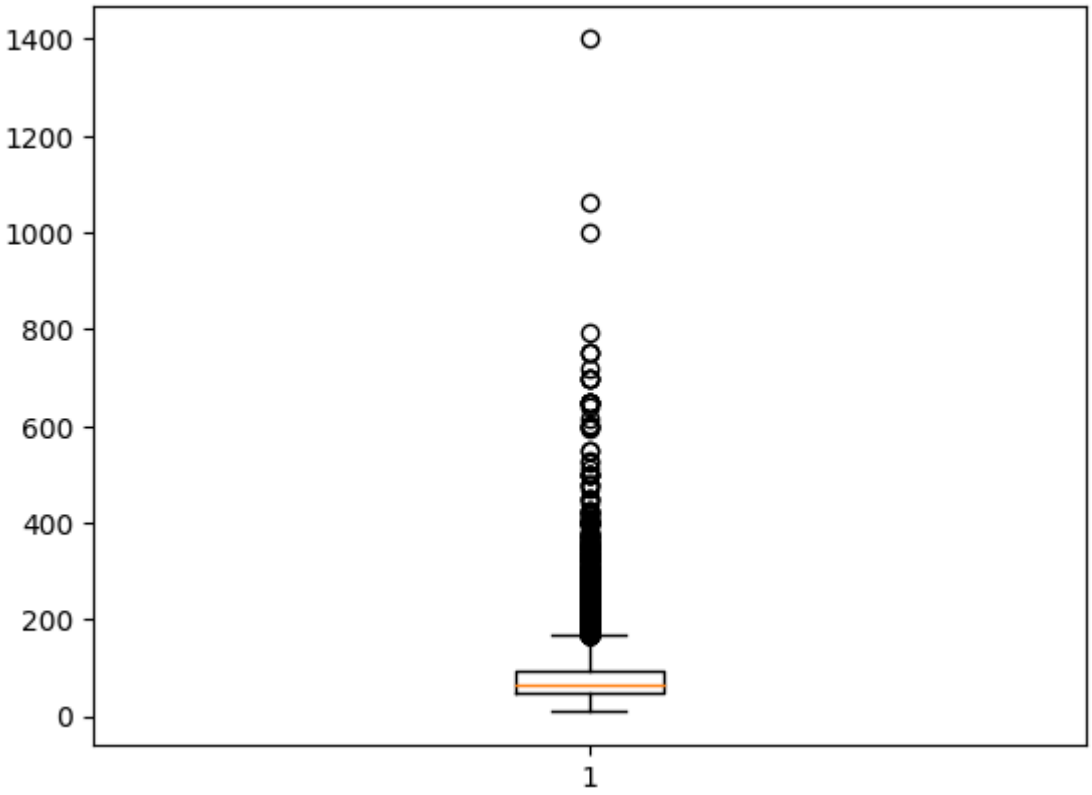

In [45]:
```
Q1 = np.percentile(df6['total_sqft'], 25.) # 25th percentile of the data of the
Q3 = np.percentile(df6['total_sqft'], 75.) # 75th percentile of the data of the
IQR = Q3-Q1 #Interquartile Range
ll = Q1 - (1.5*IQR)
```

```
ul = Q3 + (1.5*IQR)
upper_outliers = df6[df6['total_sqft'] > ul].index.tolist()
lower_outliers = df6[df6['total_sqft'] < ll].index.tolist()
bad_indices = list(set(upper_outliers + lower_outliers))
drop = True
if drop:
    df6.drop(bad_indices, inplace = True, errors = 'ignore')

plt.boxplot(df6['bath'])
plt.show()
```



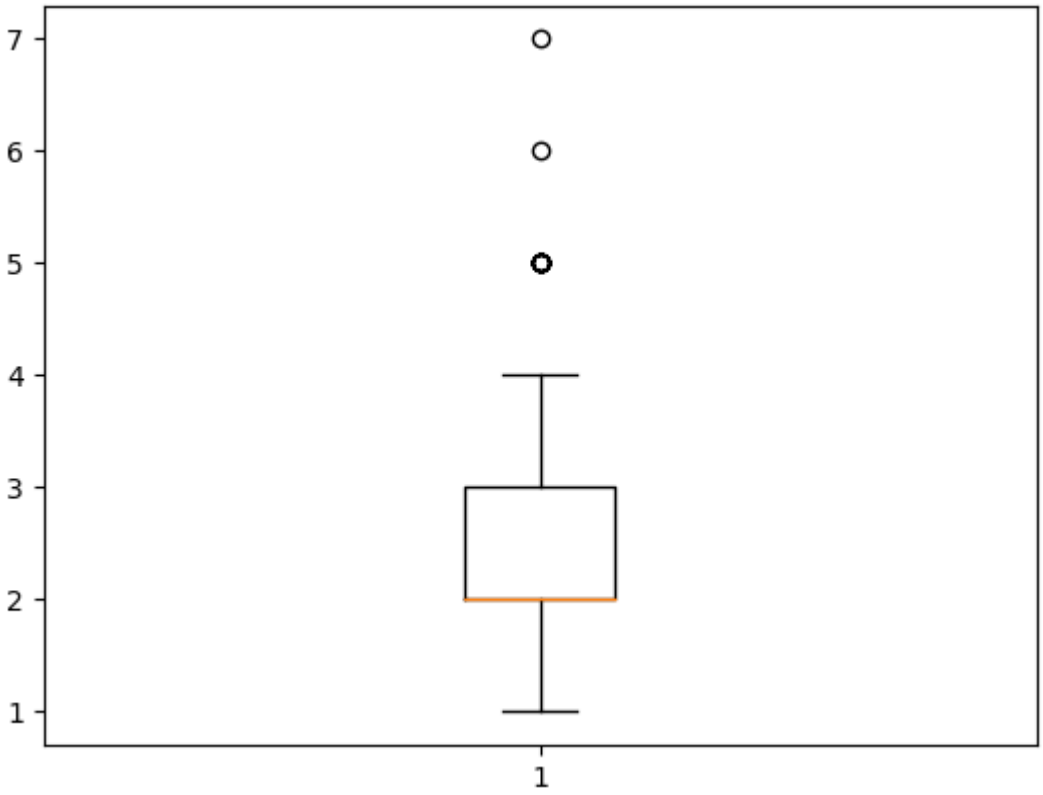

In [46]:
```
Q1 = np.percentile(df6['bath'], 25.) # 25th percentile of the data of the given
Q3 = np.percentile(df6['bath'], 75.) # 75th percentile of the data of the given
IQR = Q3-Q1 #Interquartile Range
ll = Q1 - (1.5*IQR)
ul = Q3 + (1.5*IQR)
upper_outliers = df6[df6['bath'] > ul].index.tolist()
lower_outliers = df6[df6['bath'] < ll].index.tolist()
bad_indices = list(set(upper_outliers + lower_outliers))
drop = True
if drop:
    df6.drop(bad_indices, inplace = True, errors = 'ignore')
plt.boxplot(df6['price'])
plt.show()
```




In [47]:
```
Q1 = np.percentile(df6['price'], 25.) # 25th percentile of the data of the given
Q3 = np.percentile(df6['price'], 75.) # 75th percentile of the data of the given
IQR = Q3-Q1 #Interquartile Range
ll = Q1 - (1.5*IQR)
ul = Q3 + (1.5*IQR)

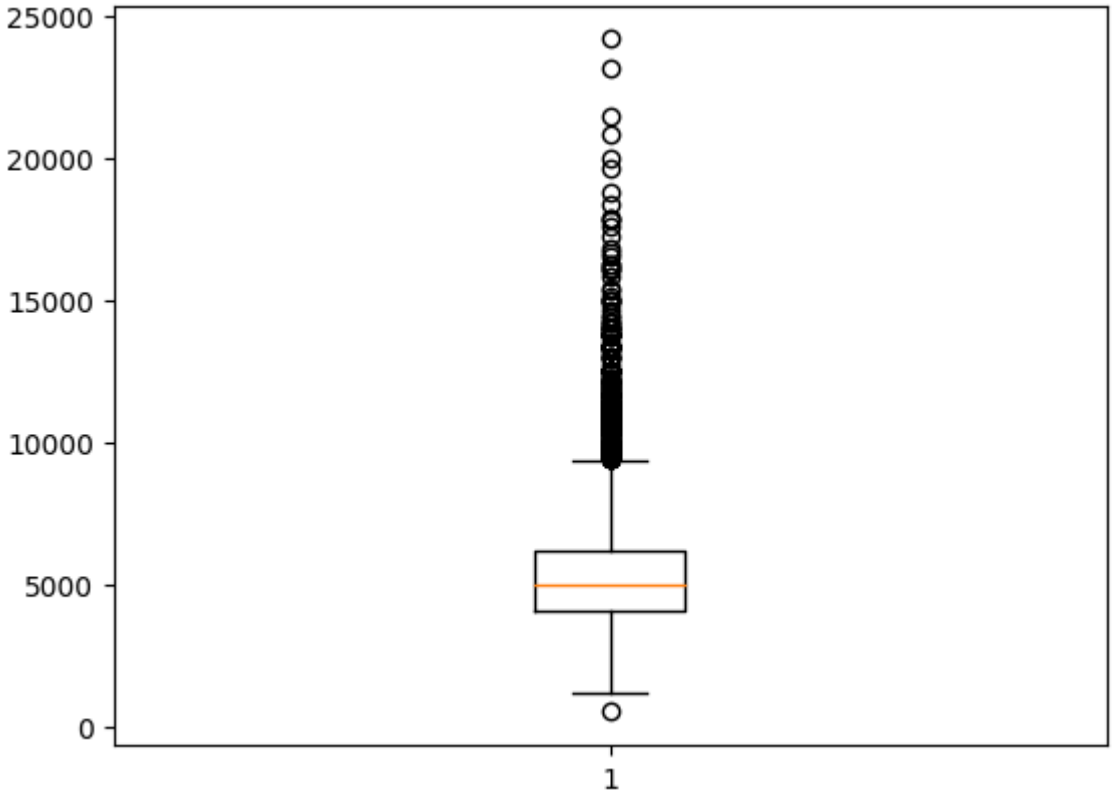
upper_outliers = df6[df6['price'] > ul].index.tolist()
lower_outliers = df6[df6['price'] < ll].index.tolist()
bad_indices = list(set(upper_outliers + lower_outliers))
drop = True
if drop:
    df6.drop(bad_indices, inplace = True, errors = 'ignore')

plt.boxplot(df6['bhk'])
plt.show()
```



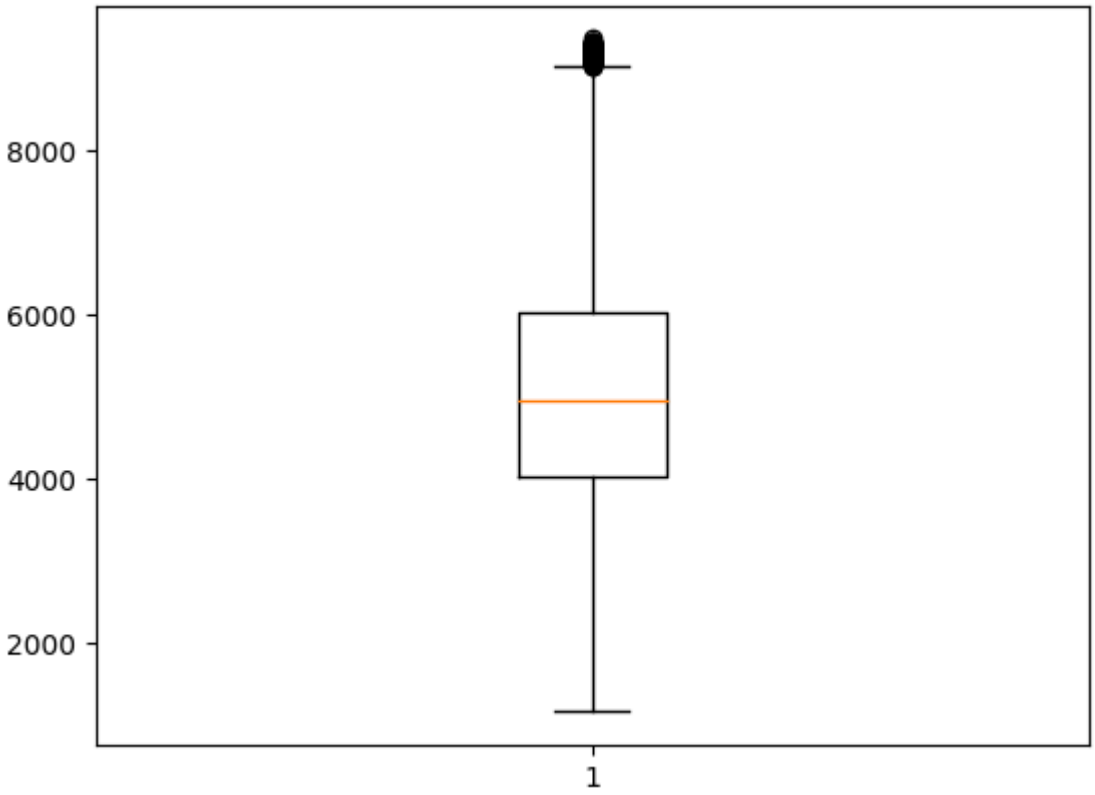

In [48]:
```
Q1 = np.percentile(df6['bhk'], 25.) # 25th percentile of the data of the given f
Q3 = np.percentile(df6['bhk'], 75.) # 75th percentile of the data of the given f
IQR = Q3-Q1 #Interquartile Range
ll = Q1 - (1.5*IQR)
ul = Q3 + (1.5*IQR)
upper_outliers = df6[df6['bhk'] > ul].index.tolist()
lower_outliers = df6[df6['bhk'] < ll].index.tolist()
bad_indices = list(set(upper_outliers + lower_outliers))
drop = True
if drop:
    df6.drop(bad_indices, inplace = True, errors = 'ignore')

plt.boxplot(df6['price_per_sqft'])
plt.show()
```




In [49]:
```
Q1 = np.percentile(df6['price_per_sqft'], 25.) # 25th percentile of the data of
Q3 = np.percentile(df6['price_per_sqft'], 75.) # 75th percentile of the data of
IQR = Q3-Q1 #Interquartile Range
ll = Q1 - (1.5*IQR)
ul = Q3 + (1.5*IQR)
upper_outliers = df6[df6['price_per_sqft'] > ul].index.tolist()
lower_outliers = df6[df6['price_per_sqft'] < ll].index.tolist()
bad_indices = list(set(upper_outliers + lower_outliers))
drop = True
if drop:
    df6.drop(bad_indices, inplace = True, errors = 'ignore')

plt.boxplot(df6['price_per_sqft'])
plt.show()
```




In [50]:
```
df6.shape
```

Out[50]:
```
(10090, 7)
```

In [51]:
```
X = df6.drop(['price'],axis='columns')
X.head(3)
```

Out[51]:

| | location | size | total_sqft | bath | bhk | price_per_sqft |
|---|---|---|---|---|---|---|
| **0** | Electronic City Phase II | 2 BHK | 1056.0 | 2.0 | 2 | 3699.810606 |
| **2** | Uttarahalli | 3 BHK | 1440.0 | 2.0 | 3 | 4305.555556 |
| **3** | Lingadheeranahalli | 3 BHK | 1521.0 | 3.0 | 3 | 6245.890861 |

In [52]:
```
X.shape
```

Out[52]:
```
(10090, 6)
```

In [53]:
```
y = df6.price
y.head(3)
```

Out[53]:
```
0    39.07
2    62.00
3    95.00
Name: price, dtype: float64
```

In [54]:
```
len(y)
```

Out[54]:
```
10090
```

In [55]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_sta
```

```
X_train.shape
```

Out[55]: (8072, 6)

In [56]:
```
y_train.shape
```

Out[56]: (8072,)

In [57]:
```
X_test.shape
```

Out[57]: (2018, 6)

In [58]:
```
y_test.shape
```

Out[58]: (2018,)

In [ ]:
```

```